

Robust Optical Pulling of Plasmonic, Chiral, and Dielectric Nanorods via Hyperbolic Metamaterials

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1. Summary:

This study explores the generation of optical pulling forces using hyperbolic metamaterials (HMMs) for advanced light-matter interaction. Plasmonic, chiral, and dielectric nanorods were analyzed under a plane-polarized laser beam using full-wave simulations. Two configurations were investigated: clustered nanorods in heterogeneous environments (water, air, and mixed) and individual nanorods in air. Results show consistent optical pulling forces on the order of 10^{-20} N for clustered systems and up to 10^{-19} N for plasmonic nanorods in single configurations. The findings reveal that plasmonic responses are highly sensitive to spatial arrangement, while dielectric behavior depends on structural dimensions. This work demonstrates a robust and scalable approach for optical manipulation, with potential applications in nanophotonics and sensing.

Keywords: *optical manipulation, pulling force, Hyperbolic Metamaterials, chirality, plasmonic.*

2. Proposed optical setup

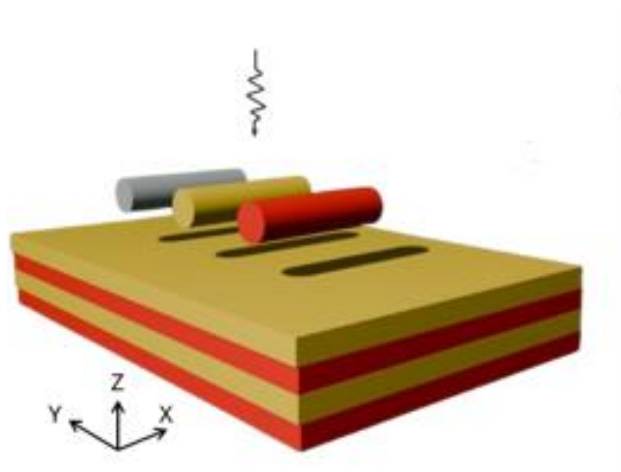


Figure 1: illustrates our proposed optical setup. It displays the 3D schematic of the clustered NRs illuminated by a non-structured time-harmonic plane-polarized wave with a magnetic field polarization along the "Y" direction, propagating from "+Z" to "-Z," and an incident angle of 0° with the normal. Three distinct color represents the corresponding three distinct materials.

3. Result:

Numerical investigations were first performed on clustered nanorod configurations, followed by single-particle analysis. Optical pulling forces were evaluated for plasmonic, chiral, and dielectric nanorods placed on a hyperbolic metamaterial (HMM) substrate under different environments (air, half air–half water, and water). Full-wave simulations reveal that all nanorods exhibit stable optical pulling forces over a broad wavelength range ($\approx 800\text{--}1600\text{ nm}$), with slight shifts depending on the surrounding medium. In clustered configurations, consistent pulling behavior is observed across all environments, while in single configurations, each nanorod independently demonstrates similar pulling characteristics. Plasmonic nanorods show strong and broadband responses, whereas dielectric and chiral nanorods

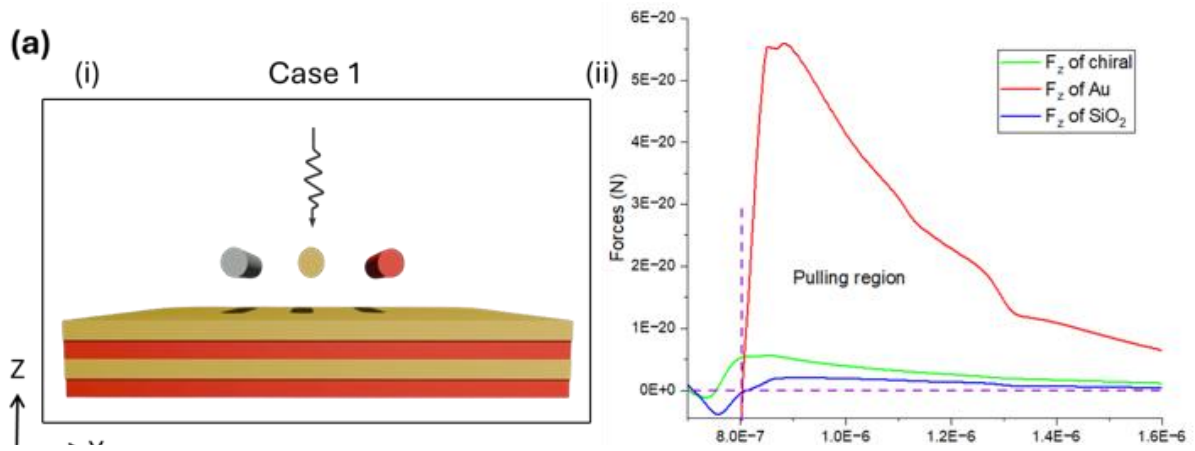


Figure 2 shows the schematic of the suggested clustered optical setup in three distinct environments, together with the accompanying full wave simulation results. Case 1 (full air medium) is illustrated in (a), the setup is schematically shown in (i), and the full-wave simulation results for our suggested Case 1 configuration are shown in (ii). Case 2 & 3 are not shown here.

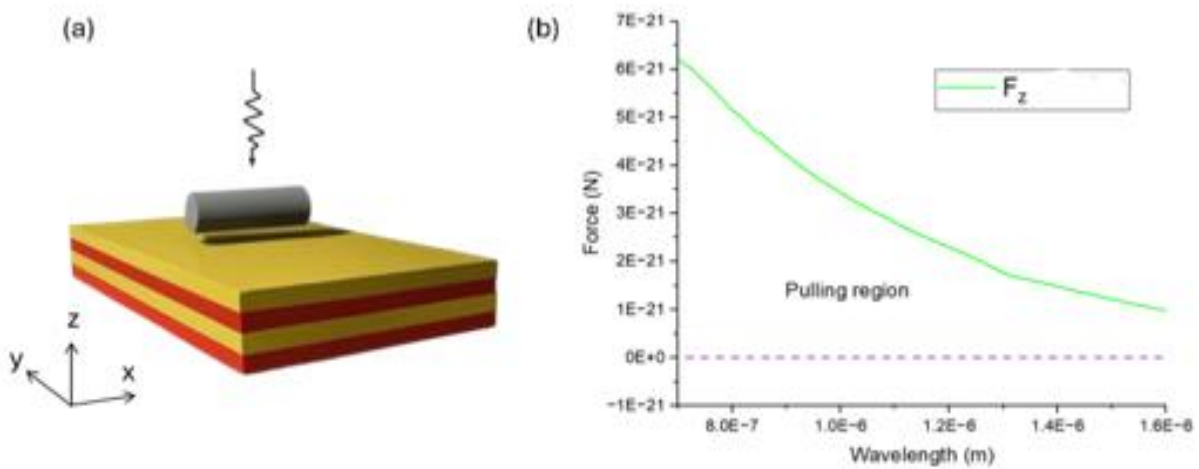


Figure 3 shows the schematic of our suggested single optical configurations (a) for schematic of single nanorod, (b) the full-wave simulation results for the corresponding nanorod.

maintain stable pulling over extended wavelength ranges. These results confirm the robustness of the proposed HMM-based optical tractor beam system for diverse configurations and environmental conditions.

To ensure a more practical and generic optical setup for getting optical pulling forces of nanorods, this research work was conducted with all possible observations as follows:

- Force Results on **Interchanging** Their Positions
- Force Results on **Nanorods to Substrate** Distances
- Force Results on **Nanorods to Nanorod** Distances
- Force Results on **Radius** Variations
- Force Results on **Length** Variations
- Force Results on **Size** Variations
- Force Results on **Without any Substrate**

To the best of our knowledge, the consistent realization of the optical tractor beam (OTB) effect across all configurations indicates the novelty and practical relevance of the proposed design.

4. Discussion:

- Mechanism of optical pulling in a chiral nanorod**

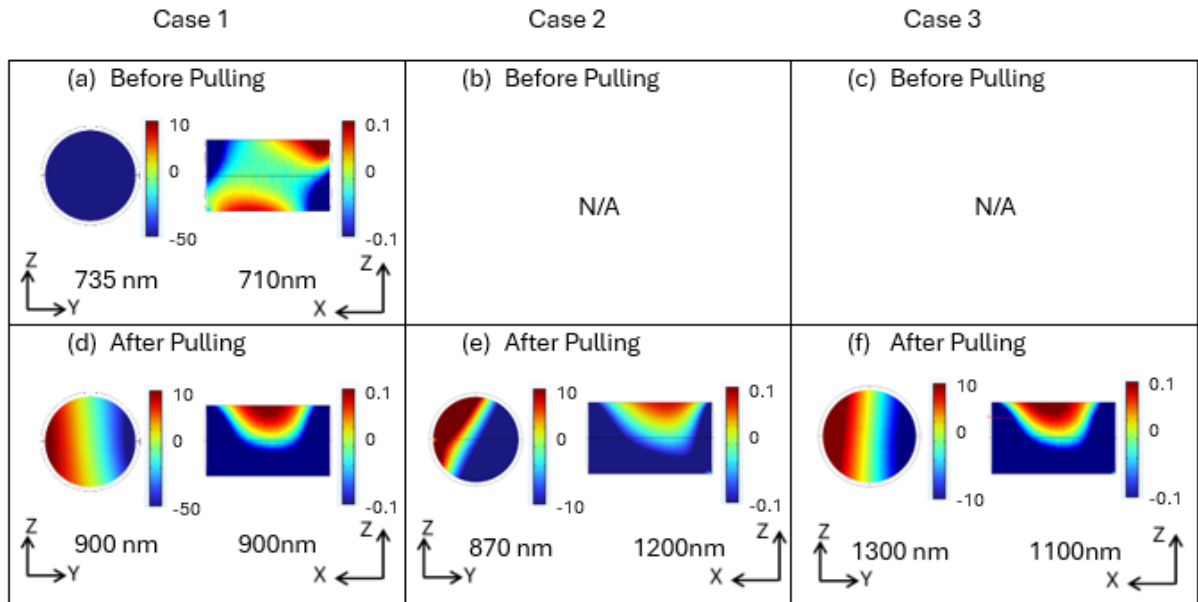


Fig. 4 shows the electric and magnetic field profile of the chiral scatterer with various incident beam wavelengths in three different environments.

b) Mechanism of optical pulling in a Plasmonic nanorod

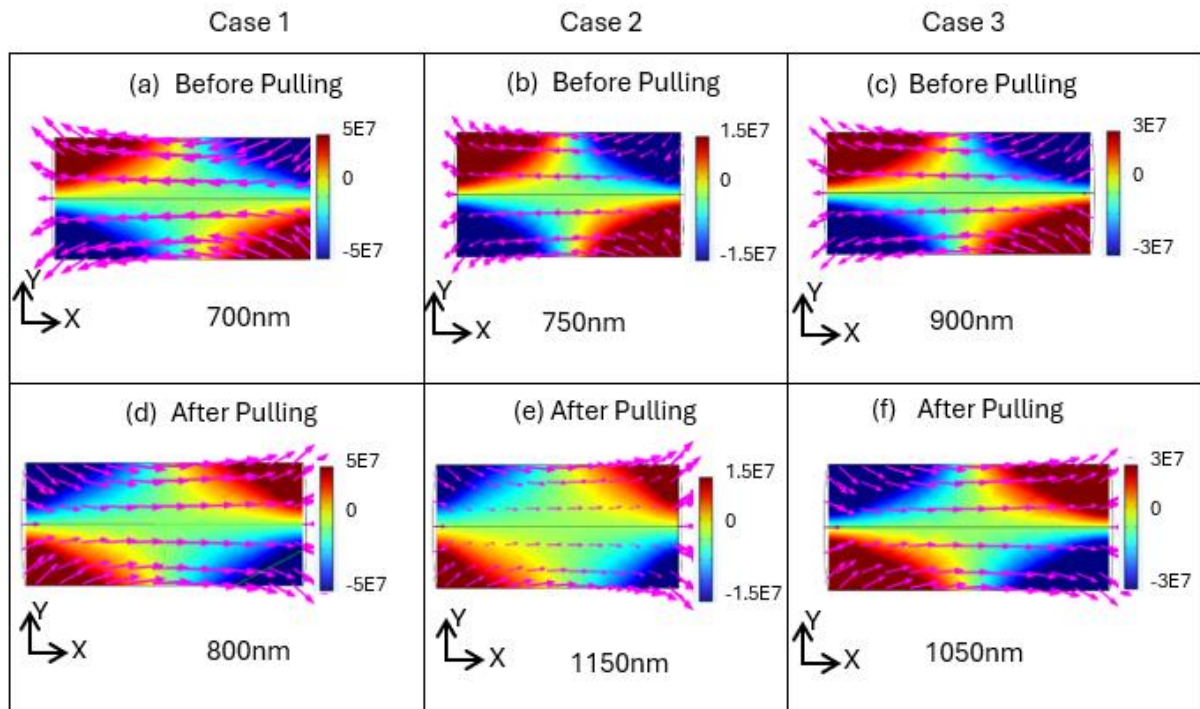


Figure 5 shows current densities (A/m²) (field and arrow profile) of plasmonic nanorod with various incident beam wavelengths in three different environments.

c) Mechanism of optical pulling in a Plasmonic nanorod

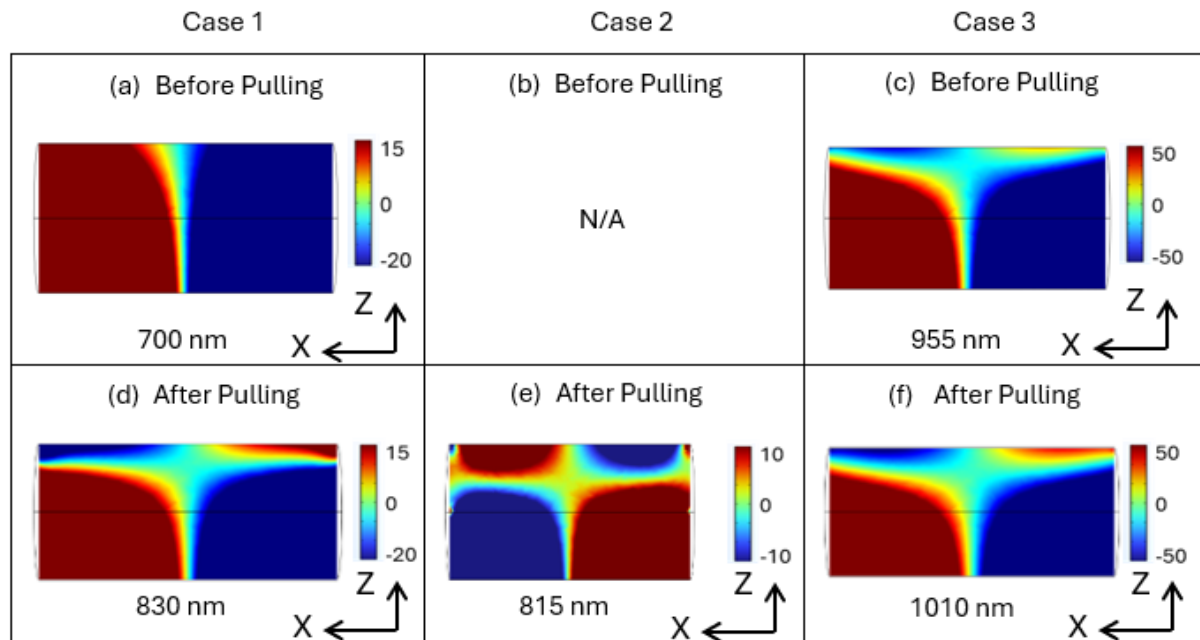


Figure 6 shows electric field distributions (V/m) of a dielectric nanorod with various incident beam wavelengths in three different environments.

5. Conclusion:

This work presents a simplified and versatile optical platform based on hyperbolic metamaterials (HMMs) for generating optical pulling forces on plasmonic, dielectric, and chiral nanorods in both single and clustered configurations. Full-wave simulations demonstrate consistent pulling forces across multiple environments, including air, half-immersed, and fully immersed water conditions. In clustered systems, optical pulling is observed over a wavelength range of 800–1600 nm, while single nanorods exhibit sustained pulling from 700–1600 nm depending on material properties. The underlying mechanisms—dipolar interference, current density reversal, and multipolar effects—are systematically analyzed. These results highlight the robustness and universality of the proposed HMM-based platform, offering strong potential for applications in nanomanipulation, photonic devices, and biosensing.

References

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